

Toronto Rush Hour Traffic Analysis*

INF2167 Project - Group 7

Stella Gregorski Yiying Qin

June 18, 2026

*Code and data are available at: https://github.com/srg-uoft/INF2167_Group7_Project.

1 Introduction

Stella: I think the goal for this is a brief 1-2 sentences to give an overview of the project

2 Research Questions(s) & Hypothesis

2.1 Questions

Stella's Rough Notes:

Note: mention using a data first approach, finding an interesting and up to date data set and determining what questions can be answered with it

What are the most accident-prone locations and times for travellers in the GTA, and how does that vary depending on mode of transportation?

Does the GTA's data align with pre-existing research on riskiest travel times?

2.2 Hypotheses

Stella's Rough Notes:

GTA traffic risk is similar to pre-established riskiest days for accidents

Some hypotheses we have are...

These are of interest because...

3 Relevant Literature

Stella's Ideas For Potential Sources

Days that have statistically the most car accidents

Mortality rates of various transportation methods

Toronto statistically has the worst traffic

congestion in Canadian cities not only costs time and sacrifices productivities, but also increases environmental burdens by more greenhouse gas emission. (<https://comt.ca/Reports/High%20Cost%20of%20Congestion%202012.pdf>)

4 Dataset & Methods

4.1 Dataset

4.1.1 Source

The data used in this analysis is the Traffic Collisions Open Data (ASR-T-TBL-001) dataset, obtained from (Toronto Police Service 2023) and licensed under the Open Government Licence – Ontario. The Toronto Police Service (TPS) maintains a Public Safety Data Portal, where they regularly publish and update datasets regarding crime statistics in the Greater Toronto Area, including the Traffic Collisions Open Data dataset.

4.1.2 Structure

TPS defines a collision for the purposes of this dataset as “the contact resulting from the motion of a motor vehicle or streetcar or its load, which produces property damage, injury or death” (Toronto Police Service 2023). To that end, each row/observation in the dataset represents a single collision, and the columns/variables describe the collision in three ways: 1) spatial/temporal information about where and when the collision occurred, 2) information about whether there were injuries, deaths, or property damage, and whether parties failed to remain at the scene, and 3) which types of vehicles were involved in the collision.

4.1.3 Key Variables

To answer the question of where the highest risk of traffic collision for various modes of transportation is in the Greater Toronto Area, our primary variables of interest are the spatial and temporal variables (to determine where and when the most accidents occur), and the information about involved vehicles (to determine any diverging trends between modes of transportation).

4.2 Methods

Methods we plan to employ include line charts for trends in accidents over time, and bar charts showing accident frequencies for each type of vehicle in each month. We also plan on utilizing summary statistic tables, particularly to show the most dangerous areas of Toronto for individual types of vehicles. This will likely involve the use of data pivoting to turn separate transportation columns into individual rows, as well as grouping and faceting to show differences in collision frequency.

5 Exploratory Data Analysis

5.1 Overview

Initial exploratory data analysis was done using statistical programming language (R Core Team 2023) and the Tidyverse library (Wickham et al. 2019), with the goal of uncovering high-level trends and possible variables of interest for further analysis. Filepaths were made relative using the Here package (Müller 2025) to improve reproducibility, and color palettes are courtesy of (Hvitfeldt 2021). Key discoveries which we hope will aid in answering our research question in future reports are elaborated on below.

5.2 Separated Temporal Variables

As mentioned in (Alexander 2023), dates can be a tricky form of data to work with, as their class and formatting can very often not work with common data analysis methods, and they can take significant cleaning to make useful. However, this dataset not only has the data represented in string format, it also has separate columns with the year and hour of the collision represented as doubles, and day of the week and month represented as character strings. Given our focus on finding the riskiest times for travellers in various transportation modes to be on the road in Toronto, the ability to count collisions for individual months or hours, as well as specific dates, will be helpful in drawing conclusions. A preliminary example is shown in (Figure 1), which shows which months have the most collisions reported.

5.3 Missing Spatial Data

An aspect of our research we may encounter more issues with is spatial data. The dataset includes a column for which Toronto neighbourhood each collision occurred in, as well as a crash latitude and longitude, which could be useful in determining where travellers face the highest risk. However, after visualizing where the most crashes are taking place, we see in (Table 1) that “NAS”, or crashes where no spatial data was recorded, is the most frequently occurring value in the Neighbourhood column, making up around 17% of the entries. We will have to determine going forward whether these should be dropped, or whether our research question could be amended to omit spatial investigation.

5.4 Types of Vehicles Stored as Columns

A key part of our analysis is investigating how risk of collision for travellers changes with mode of transportation. To this end, the TPS dataset contains 5 columns, each of which provides a yes/no as to whether cars, public transit vehicles, motorcycles, bicycles, or pedestrians were included in the accident. While this is useful data to have, it is not tidy data, and we will need

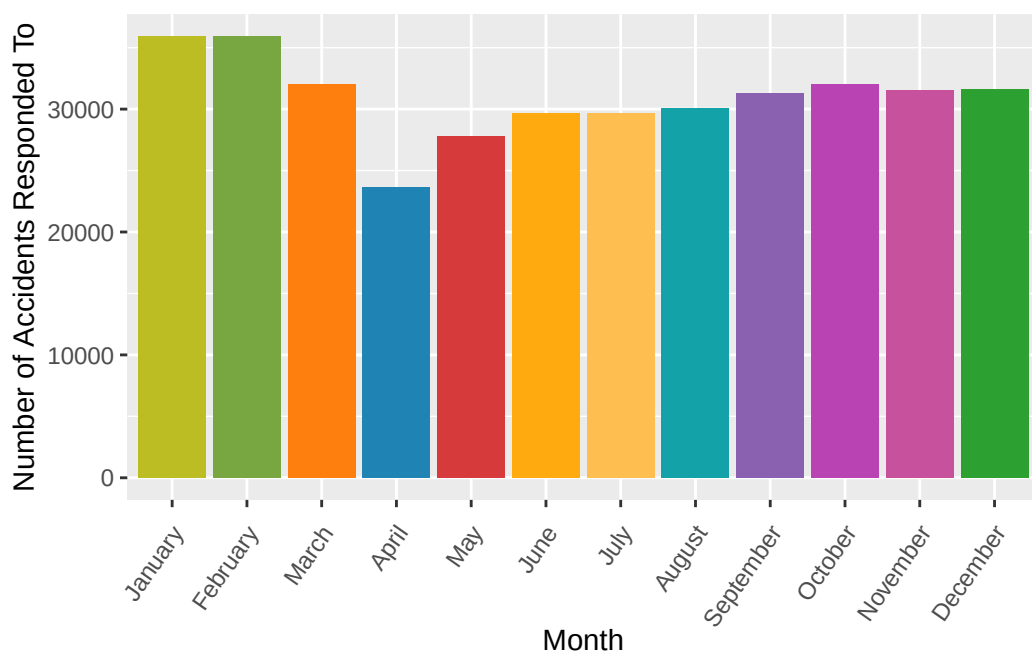


Figure 1: Accidents per Month from 2020-2026

Table 1

```
# A tibble: 159 x 3
  NEIGHBOURHOOD_158      Number_of_Collisions Percent_of_Total
  <chr>                <int>                <dbl>
1 NSA                  62076                 16.7
2 Wexford/Maryvale (119) 9853                  2.65
3 West Humber-Clairville (1) 7494                  2.02
4 York University Heights (27) 6089                  1.64
5 Dorset Park (126) 5786                  1.56
6 St Lawrence-East Bayfront-The Islands ~ 5389                  1.45
7 Milliken (130) 4988                  1.34
8 Banbury-Don Mills (42) 4870                  1.31
9 Etobicoke City Centre (159) 4775                  1.29
10 Woburn North (142) 4746                  1.28
# i 149 more rows

[1] 10
```

to pivot the dataframe so each row is an individual vehicle involved in an accident before we can determine trends by vehicle type. (Figure 2) shows a basic use case for this pivoted data, and giving us preliminary data about how frequently the various modes of transportation are involved in collisions.

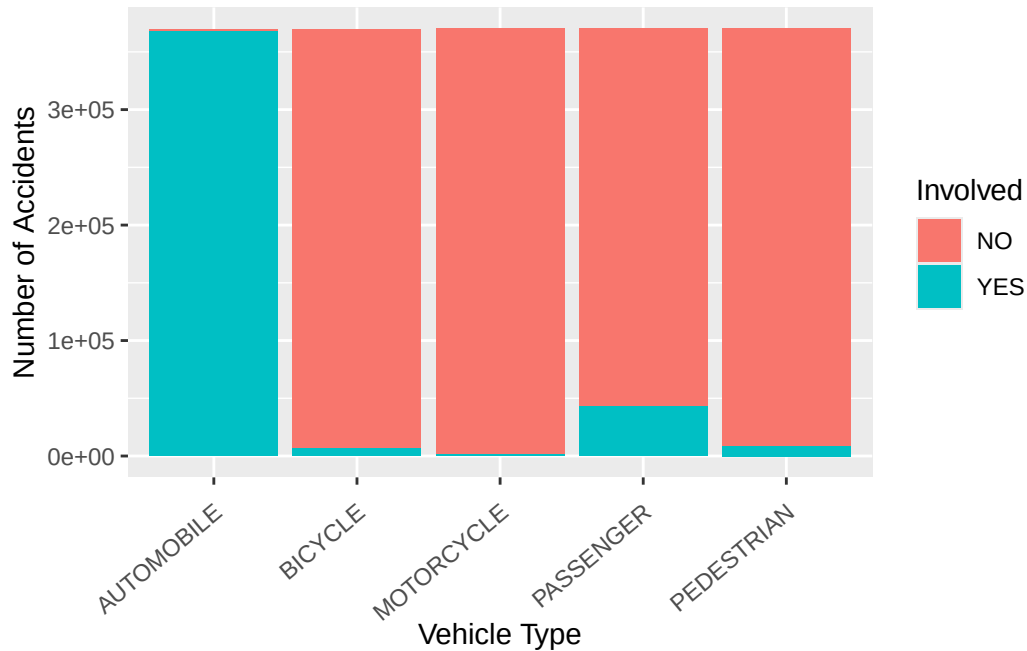


Figure 2: What Types of Vehicles Were Involved in Each Recorded Traffic Collision from 2020-2026

6 Next Steps

Having outlined the research question of interest, target data set, and preliminary exploratory data analysis above, we will move into more detailed and thorough analysis of the dataset to identify any trends which could help address the primary research question. Conclusions and discussion, along with raw results and any models used in their generation, will be presented in a final report and oral presentation in August 2026.

7 References

- Alexander, Rohan. 2023. *Telling Stories with Data*. Chapman; Hall/CRC. <https://tellingstorieswithdata.com/>.
- Hvitfeldt, Emil. 2021. *Paletter: Comprehensive Collection of Color Palettes*. <https://github.com/EmilHvitfeldt/paletter>.
- Müller, Kirill. 2025. *Here: A Simpler Way to Find Your Files*. <https://here.r-lib.org/>.
- R Core Team. 2023. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>.
- Toronto Police Service. 2023. “Traffic Collisions Open Data (ASR-T-TBL-001).” In *Public Safety Data Portal*. Toronto Police Service. <https://data.tps.ca/datasets/TorontoPS::traffic-collisions-open-data-asr-t-tbl-001/about>.
- Wickham, Hadley, Mara Averick, Jennifer Bryan, et al. 2019. “Welcome to the Tidyverse.” *Journal of Open Source Software* 4 (43): 1686. <https://doi.org/10.21105/joss.01686>.